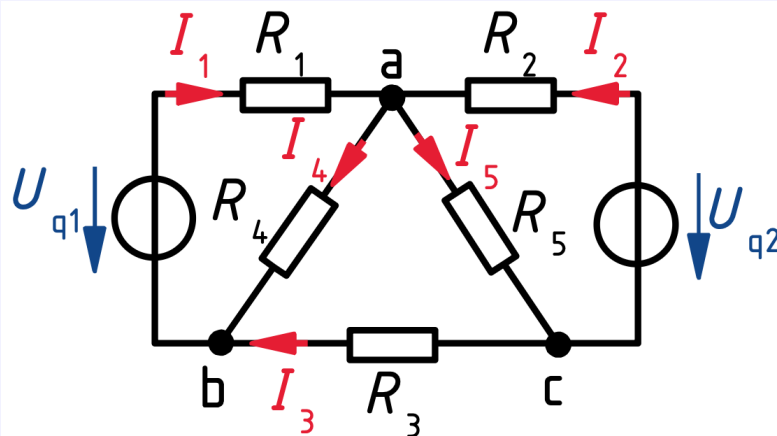


Exercise Sheet #3: DC 2: Network Analysis and Power

Discussion: Oct 29, 2025

Exercise 1: Systematic Network Analysis

The following network contains the resistors $R_1 = 1\ \Omega$, $R_2 = 2\ \Omega$, $R_3 = 5\ \Omega$, $R_4 = 25\ \Omega$, and $R_5 = 40\ \Omega$, as well as the source voltages $U_{q1} = 16.2\ \text{V}$ and $U_{q2} = 11.4\ \text{V}$. All branch currents are to be determined.



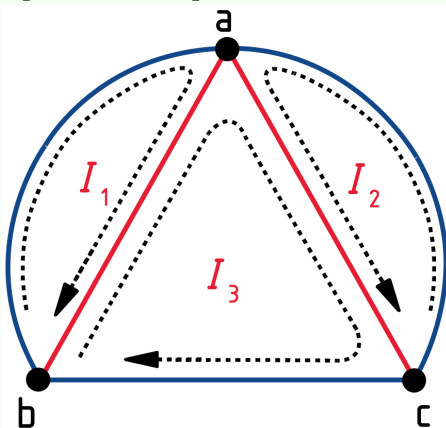
Solution

This network has $n = 3$ nodes, so there are $n - 1 = 2$ linearly independent node equations. Picking nodes B and C :

$$I_3 + I_4 - I_1 = 0$$

$$I_5 - I_2 - I_3 = 0$$

The number of branches is 5, so there are $5 - (3 - 1) = 3$ linearly independent mesh equations. Graph of the network with a selected complete tree (marked in red):



The branch currents in the connecting branches, i.e., I_1 , I_2 , and I_3 , are chosen as mesh currents. The mesh equations are set up by traversing the meshes in the direction of the

counting arrows of the mesh currents:

$$M_1 : R_1 I_1 + R_4 I_4 - U_{q1} = 0$$

$$M_2 : R_2 I_2 + R_5 I_5 - U_{q2} = 0$$

$$M_3 : R_3 I_3 - R_4 I_4 + R_5 I_5 = 0$$

Resulting system of equations as matrix equation:

$$\begin{bmatrix} -1 & 0 & 1 & 1 & 0 \\ 0 & -1 & -1 & 0 & 1 \\ R_1 & 0 & 0 & R_4 & 0 \\ 0 & R_2 & 0 & 0 & R_5 \\ 0 & 0 & R_3 & -R_4 & R_5 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \\ I_4 \\ I_5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ U_{q1} \\ U_{q2} \\ 0 \end{bmatrix}.$$

Substituting numerical values gives:

$$\begin{bmatrix} -1 & 0 & 1 & 1 & 0 \\ 0 & -1 & -1 & 0 & 1 \\ 1 & 0 & 0 & 25 & 0 \\ 0 & 2 & 0 & 0 & 40 \\ 0 & 0 & 5 & -25 & 40 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \\ I_4 \\ I_5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 16.2 \\ 11.4 \\ 0 \end{bmatrix}.$$

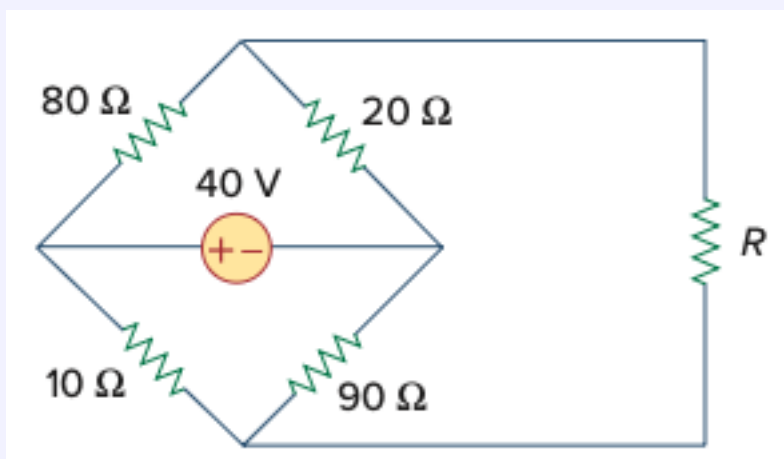
...

$$I_1 = 1.2 \text{ A}, \quad I_2 = -0.3 \text{ A}, \quad I_3 = 0.6 \text{ A}, \quad I_4 = 0.6 \text{ A}, \quad I_5 = 0.3 \text{ A}.$$

Exercise 2: Thévenin and Power Matching

The variable resistor R is adjusted until it absorbs the maximum power from the circuit.

- Calculate the value of R for maximum power.
- Determine the maximum power absorbed by R .



Solution

- a) First, we calculate the Thévenin equivalent seen from the terminals of R.

Let node D (left of 40 V source) be at 40 V, and node B (right of source) be at 0 V.

For the upper node A :

$$\frac{V_A - 40}{80} + \frac{V_A - 0}{20} = 0 \Rightarrow 5V_A - 40 = 0 \Rightarrow V_A = 8 \text{ V}$$

For the lower node C :

$$\frac{V_C - 40}{10} + \frac{V_C - 0}{90} = 0 \Rightarrow 10V_C - 360 = 0 \Rightarrow V_C = 36 \text{ V}$$

Thus, the open-circuit (Thévenin) voltage is:

$$V_{th} = |V_A - V_C| = |8 - 36| = 28 \text{ V}$$

Now we deactivate the 40 V source (replace by a short circuit):

$$R_A = (80 \Omega \parallel 20 \Omega) = \frac{80 \cdot 20}{80 + 20} = 16 \Omega$$

$$R_C = (10 \Omega \parallel 90 \Omega) = \frac{10 \cdot 90}{10 + 90} = 9 \Omega$$

Since R_A and R_C are in series,

$$R_{th} = R_A + R_C = 16 + 9 = 25 \Omega.$$

For maximum power: $R = R_{th} = 25 \Omega$

- b) The circuit can be represented by a Thévenin voltage source V_{th} in series with a Thévenin resistance R_{th} and a load R_L .

$$I = \frac{V_{th}}{R_{th} + R_L}$$

Power in the load:

$$P = I^2 R_L = \frac{V_{th}^2 R_L}{(R_{th} + R_L)^2}$$

Substitute $R_L = R_{th}$ into the power equation:

$$P_{max} = \frac{V_{th}^2 R_{th}}{(R_{th} + R_{th})^2} = \frac{V_{th}^2}{4R_{th}} = \frac{28^2}{4 \cdot 25} = \frac{784}{100} = 7.84 \text{ W}$$

Exercise 3: Efficiency

Given the efficiencies listed below, determine the corresponding fraction of the maximum load power for each case.

a) $\eta = 75\%$

b) $\eta = 90\%$

Solution

Efficiency η is defined as the ratio of power delivered to the load to the total power supplied by the source.

For a Thévenin source (V_{th} , R_{th}) with a load R_L , the efficiency is:

$$\eta = \frac{P_L}{P_L + P_{th}} = \frac{R_L}{R_L + R_{th}}$$

At maximum power ($R_L = R_{th}$):

$$P_{L,max} = \frac{V_{th}^2}{4R_{th}}$$

Hence, the power ratio is:

$$\frac{P_L}{P_{L,max}} = \frac{\frac{V_{th}^2 R_L}{(R_{th} + R_L)^2}}{\frac{V_{th}^2}{4R_{th}}} = 4R_{th} \cdot \frac{R_L}{(R_{th} + R_L)^2}$$

a) For $\eta = 75\%$:

$$\begin{aligned} 0.75 &= \frac{R_L}{R_L + R_{th}} \\ \Rightarrow R_L &= 3R_{th} \\ \Rightarrow \frac{P_L}{P_{L,max}} &= 4R_{th} \cdot \frac{3R_{th}}{(R_{th} + 3R_{th})^2} = \frac{12R_{th}^2}{(4R_{th})^2} = 0.75 \end{aligned}$$

Even when efficiency increases to 75%, the load still receives 75% of the maximum possible power. Thus, higher efficiency is achieved at the cost of only a moderate reduction in power delivered to the load.

b) For $\eta = 90\%$:

$$\begin{aligned} 0.9 &= \frac{R_L}{R_L + R_{th}} \\ \Rightarrow R_L &= 9R_{th} \\ \Rightarrow \frac{P_L}{P_{L,max}} &= 4R_{th} \cdot \frac{9R_{th}}{(R_{th} + 9R_{th})^2} = \frac{36R_{th}^2}{(10R_{th})^2} = 0.36 \end{aligned}$$

So as efficiency rises, delivered power drops sharply.

Exercise 4: Power Measurement

Calculate the effect of voltmeter and ammeter placement on the measured power in a DC resistive circuit with the parameters and configurations below. Compare the measured power to the true power dissipated in the load resistor for both configurations and for the two different load resistances.

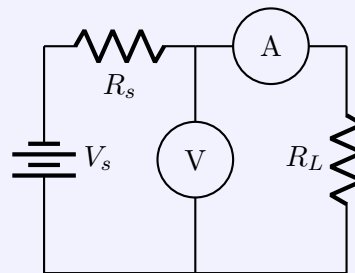
- Source voltage: $V_s = 12\text{ V}$
- Source internal resistance: $R_s = 1\ \Omega$
- Voltmeter internal resistance: $R_v = 1000\ \Omega$
- Ammeter internal resistance: $R_a = 0.1\ \Omega$

- Load resistance (two cases):

- Low load resistance: $R_L = 1\ \Omega$
- High load resistance: $R_L = 100\ \Omega$

1. Configuration A (Current-corrected):

Voltmeter connected across the source, ammeter in series with the load.



2. Configuration B (Voltage-corrected):

Voltmeter connected across the load, ammeter in series with the source.

Solution

Configuration A

$$I_s = I_L + I_v$$

$$I_L = \frac{V_t}{R_a + R_L} \quad (\text{through ammeter and load})$$

$$I_v = \frac{V_t}{R_v} \quad (\text{through voltmeter}).$$

Source terminal relation:

$$V_t = V_s - R_s \cdot I_s = \frac{V_s}{1 + R_s \left(\frac{1}{R_a + R_L} + \frac{1}{R_v} \right)}$$

Measured power (what you would calculate from the meter readings, i.e. voltmeter $\times$ ammeter):

$$P_{\text{meas},A} = V_t I_L = \frac{V_t^2}{R_a + R_L} = \frac{V_s^2}{(R_a + R_L) \left(1 + R_s \left(\frac{1}{R_a + R_L} + \frac{1}{R_v} \right) \right)^2}$$

True power dissipated in load resistor:

$$P_{\text{true}} = I_L^2 R_L = \frac{V_t^2 R_L}{(R_a + R_L)^2}$$

Configuration B

$$R_p = \frac{R_L R_v}{R_L + R_v} \quad (\text{parallel combination of load and voltmeter})$$

$$I_s = \frac{V_s}{R_s + R_a + R_p}$$

$$V_L = I_s R_p \quad (\text{voltage across load and voltmeter})$$

Measured power:

$$P_{\text{meas},B} = V_L I_s = I_s^2 R_p = \frac{V_s^2 R_p}{(R_s + R_a + R_p)^2}$$

True power dissipated in load resistor:

$$P_{\text{true}} = I_L^2 R_L$$

$$I_L = \frac{V_L}{R_L} = \frac{I_s R_p}{R_L}$$

$$\Rightarrow P_{\text{true}} = \frac{I_s^2 R_p^2}{R_L}$$

R_L	P_{true}	$P_{\text{meas},A}$	$P_{\text{meas},B}$
$1 \, \Omega$	32.62 W	35.88 W	32.65 W
$100 \, \Omega$	1.406 W	1.407 W	1.546 W